1. Convert Celsius to Fahrenheit

Write a function that converts a Celsius temperature into Fahrenheit using F = (C x 9/5) + 32.

Source Code

```
function celsiusToFahrenheit(celsius) {
    return (celsius * 9 / 5) + 32;
}

console.log(celsiusToFahrenheit(25)); // 77
```

Input and Output

Input: 25 degrees Celsius

Output: 77 degrees Fahrenheit

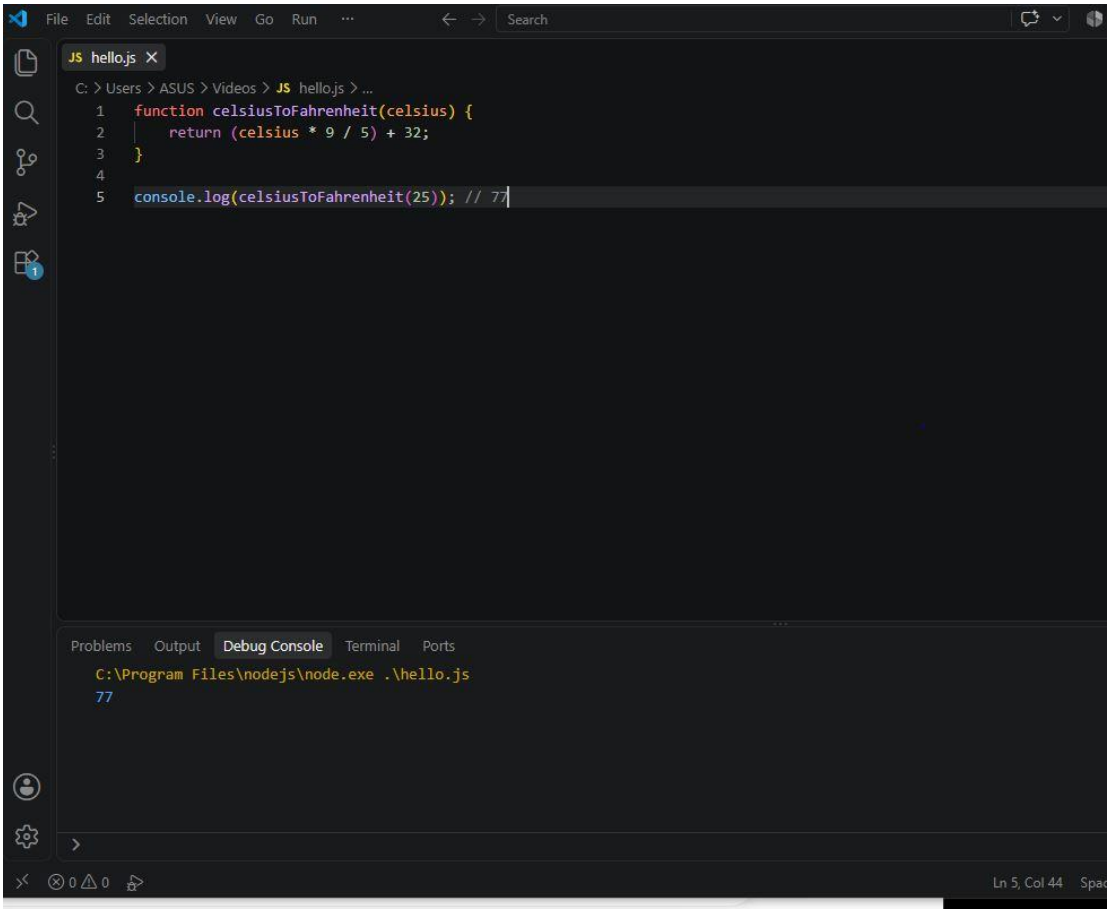


Figure: Program source code and result in Visual Studio Code

Explanation

The function receives a Celsius value as its parameter. It first multiplies the value by 9/5 and then adds 32. For an input of 25, the function returns 77, so 25 degrees Celsius is equal to 77 degrees Fahrenheit.

2. Find the Factorial of a Number

Write a function that calculates the factorial of a non-negative integer.

Source Code

```
function factorial(number) {
    if (number < 0 || !Number.isInteger(number)) {
        return "Please enter a non-negative integer.";
    }

    let result = 1;
    for (let i = 2; i <= number; i++) {
        result *= i;
    }
    return result;
}

console.log(factorial(5)); // 120
```

Input and Output

Input: 5

Output: 120

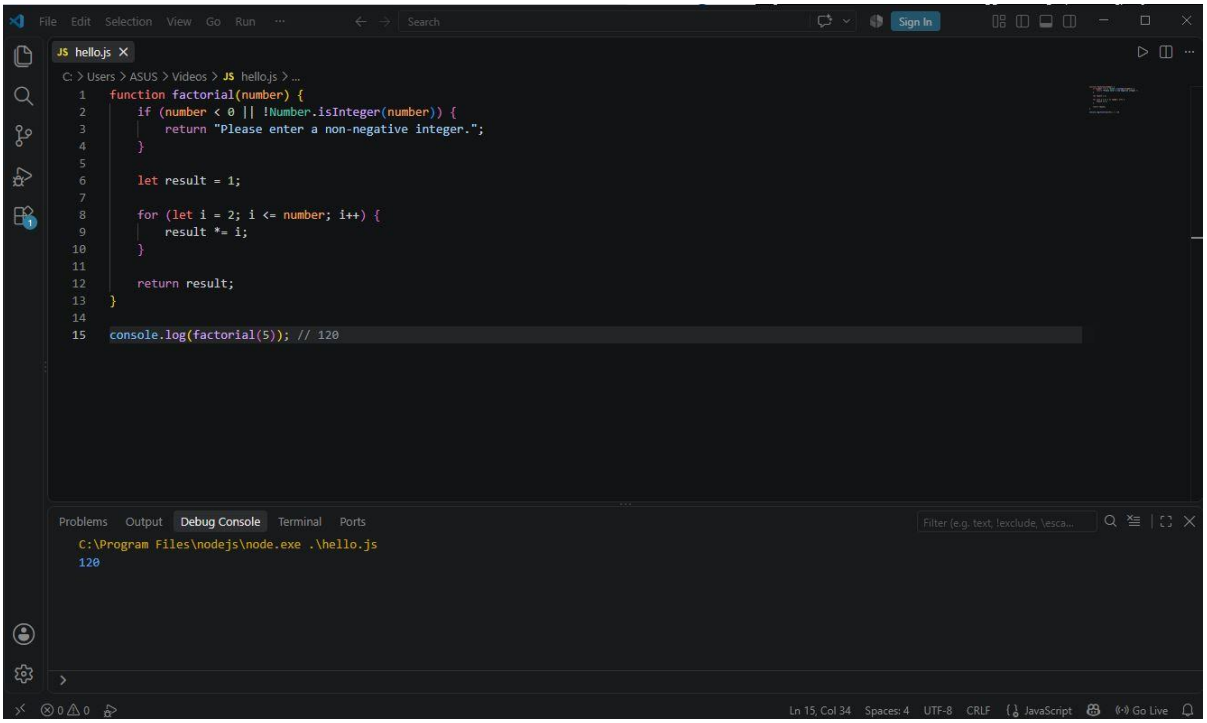


Figure: Program source code and result in Visual Studio Code

Explanation

The function first checks whether the input is a valid non-negative integer. The result starts at 1, and the loop multiplies it by every number from 2 up to the given number. Therefore, 5 factorial is calculated as 1 x 2 x 3 x 4 x 5, which gives 120.

3. Check for Palindrome

Write a function that checks whether a word reads the same forward and backward.

Source Code

```
function isPalindrome(word) {
    const reversedWord = word.split("").reverse().join("");
    return word.toLowerCase() === reversedWord.toLowerCase();
}

console.log(isPalindrome("racecar")); // true
console.log(isPalindrome("hello"));   // false
```

Input and Output

Input: "racecar" and "hello"

Output: true and false

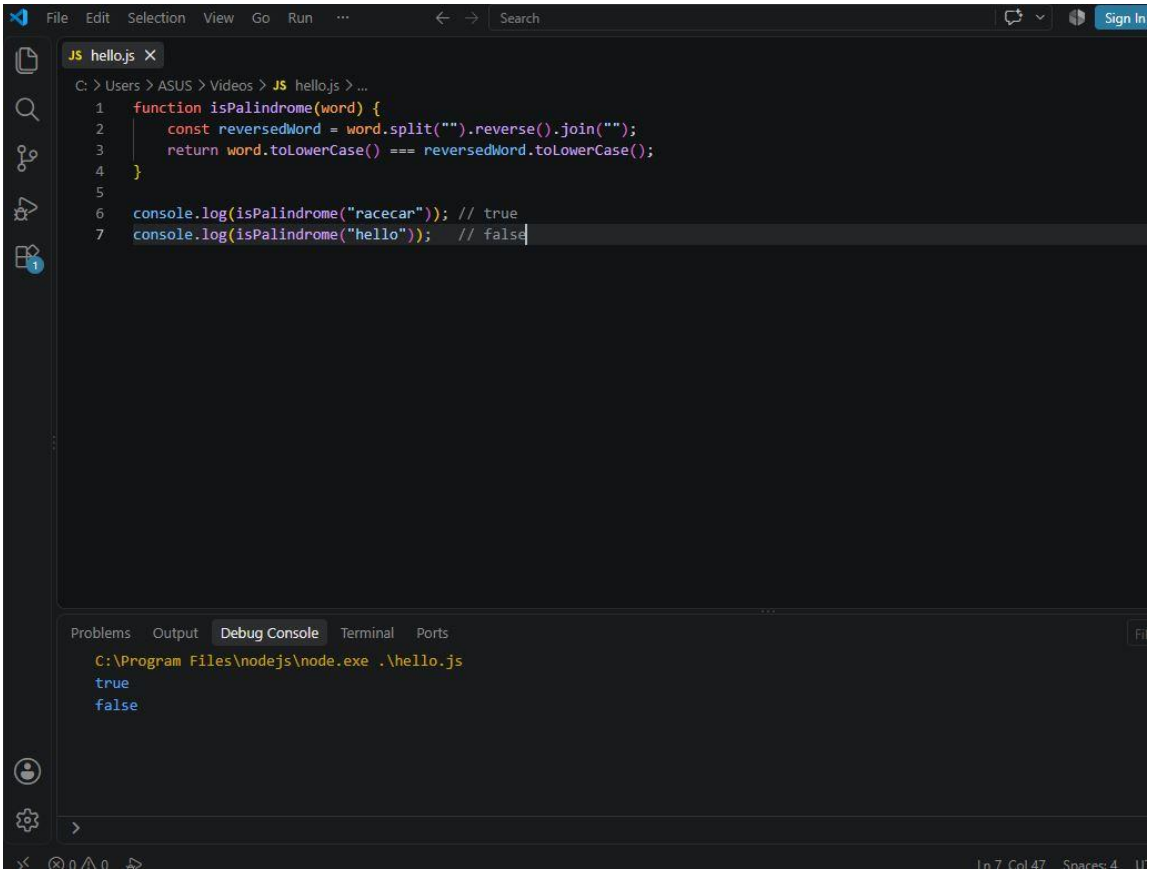


Figure: Program source code and result in Visual Studio Code

Explanation

The word is split into individual characters, reversed, and joined again. Both the original and reversed words are changed to lowercase before comparison, which makes the check case-insensitive. "racecar" is the same in both directions, while "hello" is not.

4. Sum of Array Elements

Write a function that returns the sum of all numbers in an array.

Source Code

```
function sumArray(numbers) {
    let sum = 0;

    for (const number of numbers) {
        sum += number;
    }

    return sum;
}

console.log(sumArray([10, 20, 30, 40])); // 100
```

Input and Output

Input: [10, 20, 30, 40]

Output: 100

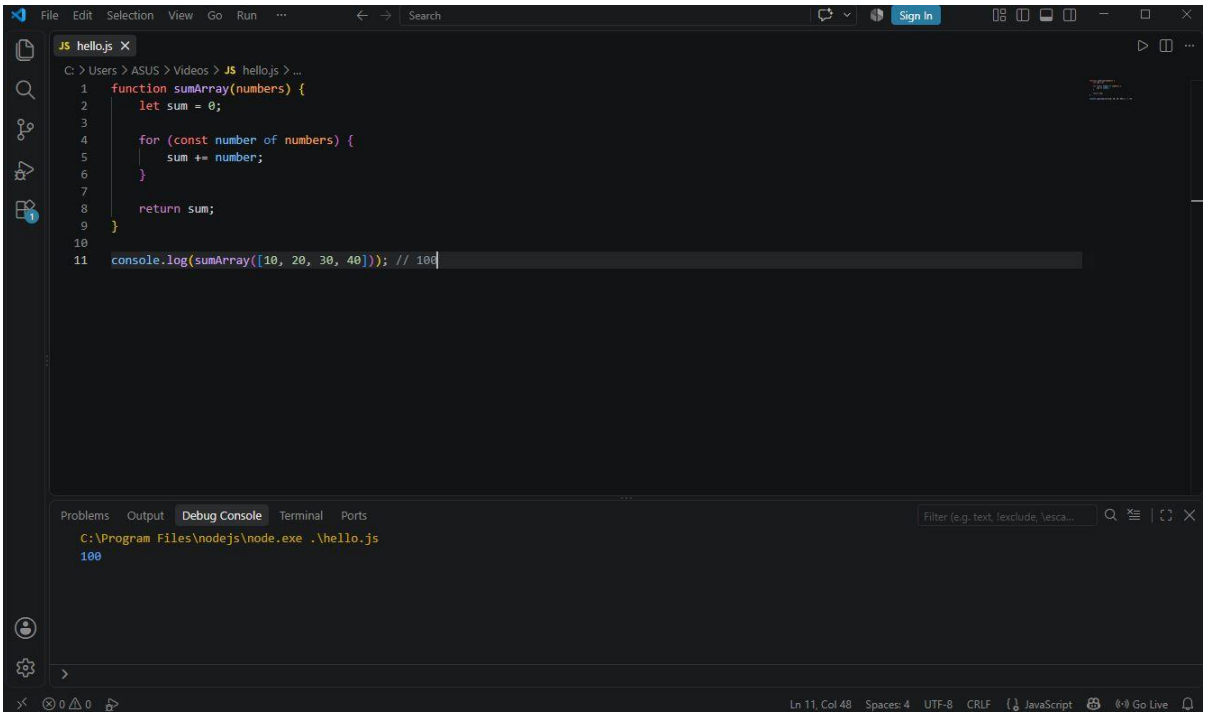


Figure: Program source code and result in Visual Studio Code

Explanation

The function starts with a sum of 0. A for...of loop visits each element in the array and adds it to the running total. After all values have been added, the function returns 100.

5. FizzBuzz Basic

Write a loop from 1 to 15 that prints Fizz, Buzz, FizzBuzz, or the number according to the divisibility rules.

Source Code

```
for (let i = 1; i <= 15; i++) {
   if (i % 3 === 0 && i % 5 === 0) {
      console.log("FizzBuzz");
   } else if (i % 3 === 0) {
      console.log("Fizz");
   } else if (i % 5 === 0) {
      console.log("Buzz");
   } else {
      console.log(i);
   }
}
```

Input and Output

Input: Numbers from 1 to 15

Output: 1, 2, Fizz, 4, Buzz, Fizz, 7, 8, Fizz, Buzz, 11, Fizz, 13, 14, FizzBuzz

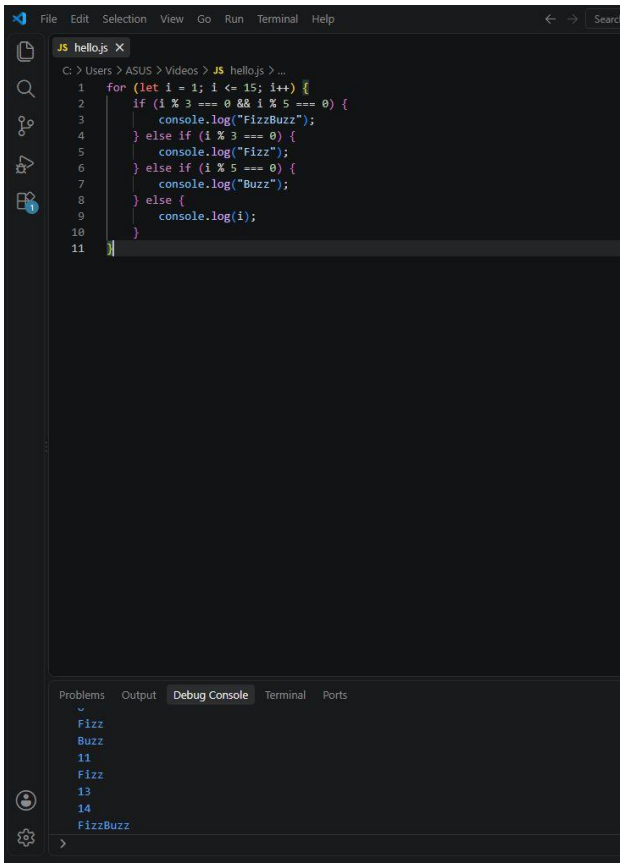


Figure: Program source code and result in Visual Studio Code

Explanation

The loop checks every number from 1 to 15. The condition for divisibility by both 3 and 5 is checked first, so 15 prints "FizzBuzz." Multiples of only 3 print "Fizz," multiples of only 5 print "Buzz," and all other values are printed normally.